RESEARCH ASSISTANT CONTRIBUTION CONFIRMATION LETTER

To Whom It May Concern,

My name is Dao Phuong Nam. I am writing this letter to confirm and acknowledge the valuable contributions made by Tran Ngoc Linh Anh in her role as a Research Assistant during high school. I had witnessed Linh Anh's abilities through her involvement in my department's research in automatic control, aerospace engineering, and artificial intelligence (AI), specifically in the development of a missile autopilot. The research project was titled "Data-Driven Q-Learning for linear systems: An approach to F16 aircraft."

Despite being a high school student, Linh Anh demonstrated a high level of professionalism and a strong analytical mindset, engaging directly with advanced research topics such as data-driven Q-learning, H-infinity robust control frameworks, and angle-of-attack-free control strategies. She contributed directly to the following aspects of the project:

- Literature Review and Dataset Preprocessing: Linh Anh was responsible for identifying and compiling reference materials and gathering technical literature on model uncertainties and flight-condition-dependent external disturbances to support the team's initial analysis.
- Simulation Support and Performance Verification: She assisted the research team in producing simulation charts for acceleration tracking, organizing evaluation tables, and cross-checking calculations under the direct guidance of the project lead. Her work supported the evaluation of the framework's effectiveness across critical flight conditions.
- Documentation and Technical Editing: She played an active role in formatting manuscripts, translating technical materials, editing document structure, and compiling regular progress reports that summarized key concepts such as Nash equilibrium learning and the Robust Integral of the Sign of the Error (RISE) feedback term.

Linh Anh's contributions helped the team work more efficiently and complete the research outputs on schedule. We greatly appreciated her dedication, ability to learn independently, and strong sense of responsibility throughout her involvement in the project.

Sincerely,
Dao Phuong Nam
Hanoi University of Science and Technology

Missile systems are inherently subject to strong nonlinearities, model uncertainties, and flight-condition–dependent external disturbances, which pose significant challenges for autopilot design. To address these issues, this paper proposes a novel angle-of-attack–free, data-driven robust control framework for missile longitudinal autopilot design, formulated as a two-player zero-sum differential game. Unlike conventional three-loop autopilots, the proposed approach does not require angle-of-attack measurements, while still achieving high-performance acceleration tracking. In addition, the method avoids the need for explicit or highly accurate system models by leveraging reinforcement learning to learn optimal control policies directly from operational data, thereby enhancing adaptability to uncertain environments. First, the acceleration tracking problem is formulated as a H∞ control problem based on the missile's dynamic characteristics, which is subsequently transformed into a zero-sum game formulation. Second, the Nash equilibrium is learned online using Q-learning based on data collected during operation. Moreover, a Robust Integral of the Sign of the Error (RISE) feedback term is incorporated to attenuate matched disturbances while requiring only knowledge of the input matrix. Simulation results validate the effectiveness of the proposed framework, demonstrating its significant improvements in robustness and performance under different complex and critical flight conditions.